

Renjie Cai

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Education

University of Electronic Science and Technology of China

School of Computer Science and Engineering, B.S. in Cyberspace Security

Aug. 2021 – Jun. 2025

Chengdu, China

- GPA: 3.81/4.0
- Selected Coursework (all with grades equivalent to 4.0/4.0): Programming Techniques Based on the Operating System, Principles of Computer Organization, Computer Architecture, Computer Network Programming, and Data Structures and Algorithms

Research Interests

Machine Learning Systems, with a focus on ML Serving, and system support for efficient inference

Research Experience

Research Assistant

Jul. 2025 – Present

University of Electronic Science and Technology of China, Advisor: Prof. Liwei Guo

- Conduct research on machine learning systems, with a focus on efficient model serving and inference system optimization.
- Worked on profiling Transformer workloads on low-end NPU chips and on serving systems for multi-robot Vision-Language-Action (VLA) workloads.

Research Intern

Sep. 2024 – Jun. 2025

University of Electronic Science and Technology of China, Advisor: Prof. Liwei Guo

- Investigated secure inference techniques for Trusted Execution Environments (TEEs), including protection of model parameters, inference data, and heterogeneous accelerator interactions.
- Designed and implemented an NPU microdriver and a TEE inference runtime to support secure execution on heterogeneous systems.

Selected Projects

Multi-Robot VLA Serving on a Single Edge GPU (*Work in Progress*)

Nov. 2025 – Present

- **Problem Setting:** Multi-robot collaboration in foreseeable real-world scenarios such as restaurants, warehouses, factories, hospitals, and hotels motivates edge VLA serving, yet robot-carried GPUs are constrained by power and compute budget, making it difficult to meet the real-time closed-loop frequency requirements of VLA control; provisioning one edge GPU per robot is prohibitively expensive, and naive multi-robot scheduling such as earliest-deadline-first reduces per-robot inference frequency and can lower task success rate during urgent task stages.
- Investigate task urgency-aware GPU compute resource spatio-temporal partitioning for edge VLA serving under real-time multi-robot workloads.

- Characterize VLA action-generation DiT as a workload distinct from conventional image/video diffusion inference, where the small action output size and limited denoising steps reduce pure compute dominance and expose other system bottlenecks.
- Observe that robot actions in VLA serving are consumed at a predictable fixed rate, enabling scheduling opportunities that are unavailable in conventional diffusion workloads with less predictable execution demand.
- Further study how increasing closed-loop control frequency can improve task success rate for urgent and highly dynamic robotic tasks.
- Design dynamic resource allocation policies across the VLM backbone and DiT stages using batching, pipelining, and scheduler-driven budget switching to improve throughput while satisfying latency constraints.

Profiling Transformer Workloads on Low-End NPU Chips

Aug. 2025 – Sep. 2025

- **Problem Setting:** As Transformers become increasingly common on edge devices, many low-end NPUs designed for traditional DNNs fail to deliver effective acceleration, leaving deployed hardware underutilized and motivating a study of how to improve Transformer inference efficiency on such platforms.
- Conducted a systematic profiling study of on-device Transformer inference on low-end NPUs, focusing on why certain workloads achieve limited acceleration or even slowdown compared with optimized CPU baselines.
- Benchmarked ViT-style workloads across three NPU platforms with progressively stronger capability, from Ethos-U65 to RK3588 and Ascend 310B, and analyzed the impact of operator support, compiler transformations, and deployment backends on end-to-end performance.
- Quantified bottlenecks such as CPU–NPU synchronization overhead, operator fallback, and memory/bandwidth-bound execution that constrained speedup on resource-limited NPU platforms.
- Used PMU-assisted and system-level profiling to study runtime cycles, bandwidth-related events, and workload characteristics for Transformer inference.

Bachelor's Thesis: TEE NPU Microdriver Design and Implementation

Nov. 2024 – Jun. 2025

- **Problem Setting:** On-device inference faces increasing privacy and model-security risks, while TEEs are primarily designed for CPUs and lack support for NPU drivers. Directly porting full NPU driver stacks into the TEE would significantly increase the trusted computing base (TCB) and introduce additional security risks, making secure NPU-accelerated inference difficult to achieve.
- Designed and implemented an NPU microdriver for Trusted Execution Environments (TEEs) on the NXP i.MX93 platform with the Arm Ethos-U65 NPU, using a record-and-replay approach to achieve equivalent NPU driver functionality with a compact trusted computing base (TCB) and without porting the full software stack.
- Built a TEE inference runtime system that supports end-to-end NPU inference with confidentiality and integrity guarantees.
- Validated correctness using multiple inference models and input datasets, and demonstrated advantages over the original driver in startup latency, memory footprint, and security.
- Designed and implemented a heterogeneous confidential inference system that coordinates the A-core and M-core for secure data loading, NPU inference, and result delivery.

Selected Systems Course Projects

Before Sep. 2024

- Completed selected systems-oriented course projects from Berkeley CS 61A/61B/61C, MIT 6.1810, and CMU 15-213.
- Built a strong foundation in computer systems and operating systems through implementation-intensive coursework.

Technical Skills

Languages and Systems: C, Java, Python, Linux Command Line, Git, Vim, GDB, MCU and Linux Kernel Driver Development

Tools: Coding Agents, AI-assisted Development Tools